

Bioinspired Silicone-Epoxy Coatings: Combining Cycloaliphatic Resins and Tetrasulfide Bonds for Enhanced Anticorrosion and Antifouling Performance

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Cite This: *Ind. Eng. Chem. Res.* 2025, 64, 5805–5817

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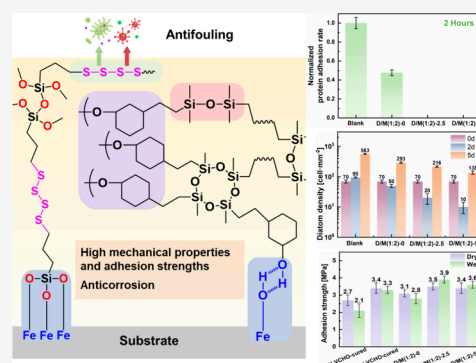


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ABSTRACT: Marine biofouling and seawater corrosion present significant economic challenges to the marine industry. While polymer coatings can effectively delay these processes, traditional coatings often release harmful substances, leading to environmental contamination. Silicone-based fouling release coatings (FRCs) improve antifouling performance by minimizing adhesion with marine organisms, but their low mechanical strength and poor substrate adhesion limit their broader application. This study combines the mechanical properties of cycloaliphatic silicone-epoxy resins with the antimicrobial capabilities of the tetrasulfide (S–S–S–S) bonds in bis(γ -triethoxysilylpropyl)tetrasulfide (Si-69) to address these limitations. Under UV irradiation, cycloaliphatic silicone-epoxy resins form a cross-linked network, ensuring excellent mechanical strength and strong substrate interactions. Simultaneously, Si-69 forms silicon–oxygen–metal (Si–O–M) bonds while maintaining the stability of the S–S–S–S bonds within the coating. The resulting UV-curable coating demonstrated superior antifouling and anticorrosion performance. It achieves a maximum tensile strength of 18.5 MPa, a hardness of 4 H, and strong adhesion to substrates under both dry (3.5 MPa) and wet (3.9 MPa) conditions. The antimicrobial activity of the S–S–S–S bonds enhances the resistance to protein and diatom adhesion. These results demonstrate the potential of this coating for high-performance FRCs in marine environments and its applications in wastewater treatment facilities, heat exchangers, and biomedical devices.



1. INTRODUCTION

The marine industry plays a key role in global economic development and transportation but faces ongoing challenges from marine biofouling and seawater corrosion, which cause considerable economic losses.¹ Research indicates that polymer coatings with antifouling and anticorrosion properties effectively reduce the adhesion of marine organisms and mitigate seawater-induced corrosion on ships and marine equipment. Traditional antifouling and anticorrosion coatings typically incorporate antifouling agents into paint films, which gradually release these agents to prevent marine organism adhesion. However, this method results in environmental pollution and does not comply with current environmental standards.² As a result, the development of nontoxic, efficient, and environmentally friendly antifouling and anticorrosion coatings with long-term stability has become a major focus of research.

In recent years, fouling release coatings (FRCs) based on the low surface energy, elasticity, and flexibility of polymer chains have gained significant attention. These coatings reduce the adhesion strength of marine organisms, enabling their detachment under the action of water flow, thus providing antifouling and anticorrosion protection for ships and marine equipment. Unlike traditional coatings that rely on the gradual

release of antifouling agents, FRCs do not release harmful substances into the environment, addressing pollution issues at the source.³ Consequently, FRCs have become a critical focus in the development of environmentally friendly and efficient marine antifouling coatings. FRCs polymers are primarily categorized into silicone-based coatings, fluorinated coatings, and fluorocarbon-modification coatings. Among these, silicone-based FRCs, particularly polydimethylsiloxane (PDMS), have garnered significant attention due to their exceptional properties, including low surface energy, low elastic modulus, low surface roughness, and long-term antifouling performance.⁴ However, despite these advantages, silicone-based FRCs are hindered by inherent limitations such as poor adhesion to substrates and insufficient mechanical strength, which restrict their wider application and further advancement in the field. To address these issues, two main strategies have been employed:⁵ (a) incorporating inorganic particles, such as

Received: November 24, 2024

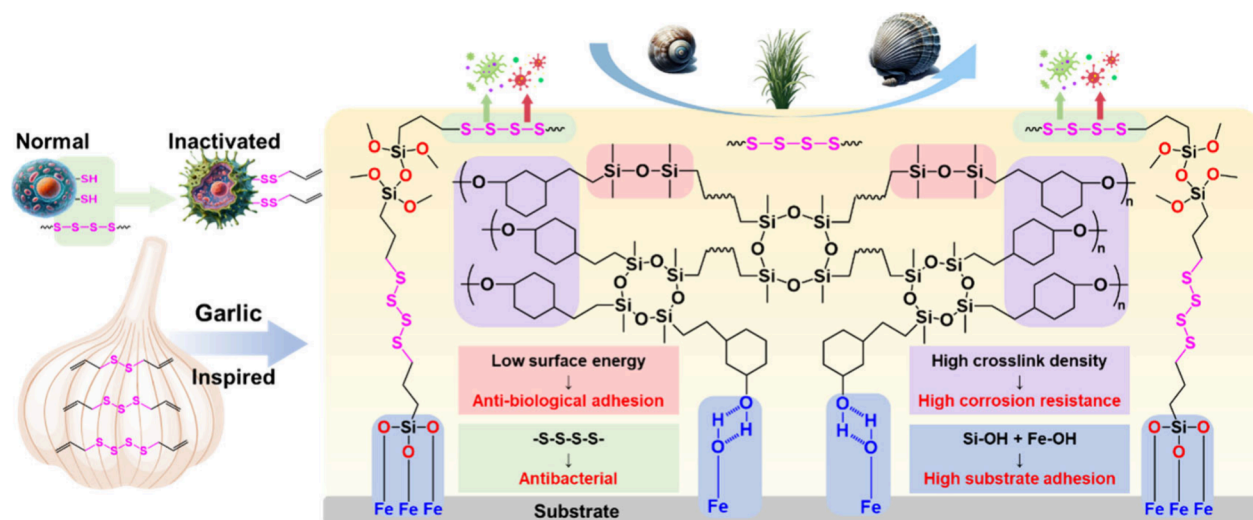
Revised: February 19, 2025

Accepted: March 3, 2025

Published: March 11, 2025



Scheme 1. Design and Mechanism of Silicone-Epoxy Coating Featuring S–S–S–S Bonds in the Si-69 Structure



SiO_2 ,^{6,7} and TiO_2 ,^{8,9} to enhance mechanical properties, and (b) chemically modifying silicone polymer.⁴ Modifications involving the introduction of epoxy resins,^{10,11} polyurethane,^{12–14} or poly(methacrylate)^{15–17} improve molecular polarity, structural rigidity, and cross-linking density, thereby enhancing both mechanical strength and adhesion to substrates. However, these approaches often compromise antifouling performance, either by increasing surface free energy or by increasing surface roughness. Therefore, improving both the mechanical strength and the adhesion to substrates of silicone-based FRCs, while simultaneously maintaining their excellent antifouling performance, remains a significant challenge in the field.

The aforementioned challenge means that a coating must simultaneously exhibit two opposing properties: high adhesion strength to substrates and low adhesion strength to marine organisms. Fortunately, we discovered an approach to reconciling these two requirements. By integration of two structurally similar components into a single resin, it is possible to achieve distinct functions within one coating. Specifically, one component provides high mechanical strength and strong adhesion to the substrate, while the other enhances the antifouling performance through its bioactivity. This dual-function strategy enables the resin to simultaneously meet both the mechanical and antifouling demands.

Research shows that cycloaliphatic silicone-epoxy resins synthesized from 1,3,5,7-tetramethylcyclotetrasiloxane (D_4H) exhibit superior mechanical properties and toughness, as they combine the rigidity of Si–O four-membered rings with the flexibility characteristic of siloxane chains.^{6,18–21} Additionally, they possess four epoxy functional groups, and when used in combination with other silicone-epoxy resins containing two epoxy groups, they enable the modulation of mechanical strength while minimizing the impact on surface free energy. Furthermore, they exhibit cationic photopolymerization capability, with a photopolymerization rate higher than that of aliphatic epoxy resins.²² Therefore, the development of D_4H -based cycloaliphatic silicone-epoxy resins holds great potential for addressing the issues of poor mechanical strength and substrate adhesion in traditional silicone resins while simultaneously advancing the application of UV-curing technology in large-scale marine industrial equipment and

fostering the creation of new environmentally friendly antifouling and anticorrosion resins.

Garlic, commonly used as a food seasoning, has also been employed since ancient times to treat bacterial infections. Its active antimicrobial component, garlic oil (GO), is composed of aliphatic polysulfides, including diallyl sulfides and diallyl trisulfide. GO exhibits inhibitory effects on a wide range of microorganisms, such as *Staphylococcus aureus*, *Escherichia coli*, and *Mycobacterium tuberculosis*.^{9,10,23,24} The primary mechanism of its antibacterial activity involves the polysulfide bonds in these compounds reacting with free thiol (–SH) groups on bacterial membrane proteins, forming disulfide bonds.^{25,26} This process disrupts the membrane structure, thereby interfering with bacterial metabolism and ultimately leading to cell death. Inspired by the antibacterial mechanism of garlic oil (GO), we speculate that bis(γ -triethoxysilylpropyl)-tetrasulfide (Si-69), a silane coupling agent with S–S–S–S bonds,^{7,8,27,28} may also exhibit significant antimicrobial activity. This idea is based on the structural similarity of the polysulfide bonds in Si-69 to those in GO, which play a critical role in disrupting bacterial membranes. Notably, the structure of Si-69 is similar to that of cycloaliphatic silicone-epoxy molecules, as both contain Si–O bonds. This structural similarity allows Si-69 to be uniformly dispersed in cycloaliphatic silicone-epoxy resins based on compatibility. Additionally, the antimicrobial activity of Si-69 enhances the antifouling capabilities of the resins, supporting their application in marine antifouling and anticorrosive coatings.

In this study, we leverage the structural similarity between cycloaliphatic silicone-epoxy resins and Si-69 to develop a UV-curable epoxy coating that combines the high mechanical strength of epoxy resins with the antimicrobial properties of Si-69. The resin, composed of tetraepoxy cyclohexanethyl 2,4,6,8-tetramethylcyclotetrasiloxane ($\text{D}_4\text{H-VCHO}$), diepoxycyclohexanethyl 1,1,3,3-tetramethyldisiloxane (MMH-VCHO), and Si-69, forms a highly cross-linked silane network through epoxy ring-opening under the action of a photoinitiator. Under UV irradiation, $\text{D}_4\text{H-VCHO}$ and MMH-VCHO react to form a cross-linked network, where the polar functional groups in the structure establish strong interactions with the substrate surface through hydrogen bonding. Meanwhile, Si-69 undergoes hydrolysis and forms silicon–oxygen–metal (Si–O–M)

bonds with the substrate surface (Scheme 1). The synergistic effect of these two mechanisms imparts the coating with excellent mechanical properties and corrosion resistance. Additionally, the low surface energy of Si–O bonds, combined with the antimicrobial activity of S–S–S–S bonds within Si-69, provides superior antifouling capabilities. This innovative UV-curable silicone-epoxy coating achieves a tensile strength of 18.5 MPa and a hardness of 4 H and demonstrates adhesion strengths of 3.5 MPa under dry conditions and 3.9 MPa under wet conditions. It also exhibits exceptional resistance to both protein and diatom adhesion, making it an ideal solution for marine antifouling and anticorrosion applications.

2. EXPERIMENTAL SECTION

2.1. Materials. 1,3,5,7-Tetramethylcyclotetrasiloxane (D₄H) was commercially obtained from Shuangtao Fine Chemical Co., Ltd. (Guangzhou, China). 1,1,3,3-Tetramethyldisiloxane (MMH) was purchased from OSBANG New Material Co., Ltd. (Shenzhen, China). 1,2-Epoxy-4-vinyl cyclohexane (VCHO) was provided by Taitel New Material Technology Co. Ltd. (Jiangsu, China). Bis-[γ -(triethoxysilyl)-propyl]tetrasulfide (Si-69) was purchased from Yuanda New Material Co., Ltd. (Guangzhou, China). Karstedts catalyst solution was supplied by Platinum Realm New Materials Co. Ltd. (Shanghai, China). Diphenyl(4-phenylthio)-phenylsulfonium hexafluoroantimonate (trade name, PI-PAG-002) and 2-isopropylthioxanthone (trade name, PI-ITX) were provided by Yangfan New Material Co. Ltd. (Zhejiang, China). All of the materials were used as received. Deionized water was used to prepare all of the aqueous solutions in this work.

2.2. Synthesis of Tetraepoxy Cyclohexanethyl 2,4,6,8-tetramethylcyclotetrasiloxane (D₄H-VCHO, Figure S1a). 52.00 g (0.42 mol) portion of VCHO, 52.00 g of toluene, 0.03 g (33.40 ppm) of Karstedts catalyst solution, and 0.25 g of anhydrous ethanol were added to the four-neck round-bottom flask equipped with a serpentine condenser, stirring rod, thermometer, and constant pressure funnel. The temperature of the reaction mixture was raised to 80 °C with an oil bath, and 24.00 g (0.1 mol) of D₄H was added dropwise to maintain the temperature at 80 °C. After completion of the reaction, the synthetic product was distilled under reduced pressure at 70 °C and –0.1 MPa for 3 h to remove toluene and excess reactants and then cooled to room temperature to obtain a colorless transparent liquid D₄H-VCHO.²²

2.3. Synthesis of Diepoxycyclohexanethyl-1,1,3,3-tetramethyldisiloxane (MMH-VCHO, Figure S1b). 330.00 g (2.6 mol) of VCHO, 330.00 g of toluene, 0.063 g (10 ppm) of Karstedts catalyst solution, and 1.50 g of anhydrous ethanol were added to a four-neck round-bottom flask equipped with a serpentine condensation tube, a stirring rod, a thermometer, and constant pressure funnel. The temperature of the reaction mixture was raised to 80 °C with an oil bath; then 176.80 g (1.30 mol) of MMH was added dropwise at 70–80 °C, and the mixture was stirred at 60 °C for 2 h after the titration was completed. After the reaction was completed, vacuum distillation was carried out at 70 °C and –0.1 MPa for 3 h to remove toluene and excess reactants, and the colorless transparent liquid MMH-VCHO was obtained after cooling to room temperature.²⁹

2.4. Preparation of Photocurable Cycloaliphatic Silicone-Epoxy Coatings (D/M(*m*/*n*)-*x*). Photoinitiator PAG-002 (PI-PAG-002) and photoinitiator ITX (PI-ITX) were mixed in a ratio of 6:1 and marked as PI-mix. Weigh PI-

mix (the amount of cationic photoinitiator is 2% of the total prepolymer weight) and dissolve it in acetone. D₄H-VCHO and MMH-VCHO are mixed in a certain weight ratio, yielding the product as transparent resin, which is designated as D/M(*m*:*n*) (*m*:*n* = 1:1, 1:2, 2:1) (where *m*:*n* refers to the mass ratio of D₄H-VCHO and MMH-VCHO). For example, 1/1 represents that the mass ratio of D₄H-VCHO to MMH-VCHO is 1:1. The mixture is mechanically agitated and mixed uniformly while avoiding light. Then the mixture mentioned above is divided into two vials and silane coupler Si-69 is added in different proportions (the amount of silane coupler is 0%, 2.5%, and 5% of the total weight of the prepolymer, respectively). The prepolymers above were designated as D/M(*m*:*n*)-*x* (*x* = 0, 2.5, and 5), where 0, 2.5, 5 refer to the mass ratio of Si-69 in the D/M(*m*:*n*). Then the mixture was mechanically stirred to mix evenly and ultrasonically stirred for 15 min to remove suspended air bubbles. Finally, we use the mixture to prepare a coating film of a certain thickness and expose it to 365 nm ultraviolet light curing for 10 min at room temperature to obtain the silicone coating, the structure of cross-linked network for the mixture of D₄H-VCHO and MMH-VCHO as shown in Figure S1c.

2.5. Characterization. Fourier transform infrared (FT-IR) spectrum was measured using a Nicolet iS50 spectrometer (Thermo Scientific, Germany). The samples were prepared as KBr pellets and recorded from 4000 to 400 cm^{–1} with a resolution of 4 cm^{–1}. Real-time infrared spectra (RT-IR) were employed to estimate the photopolymerization conversion of silicone-epoxy in resin (*C_t*) using a Nicolet iS50 spectrometer (Thermo Scientific) with KBr pellets. The silicone-epoxy prepolymer was irradiated in situ using UV-LED light source with a wavelength of 365 nm (LAMPLIC, Shenzhen), and the irradiation intensity was measured by UV light meter (LS125, 365 nm detector, Shenzhen Lin-Shang Technology Co., Ltd.). The *C_t* was calculated from the variation of the epoxy group absorption peak at 883 cm^{–1} according to eq 1,^{30,31}

$$C_t = \frac{(A_0 - A_t)}{A_0} \times 100\% \quad (1)$$

where *A*₀ and *A_t* were the peak area of silicone-epoxy at 883 cm^{–1} before and after UV irradiation for different times, respectively.

¹H NMR and ²⁹Si NMR spectra were recorded using a Bruker AVANCE III HD 400 MHz superconducting Fourier spectrometer (Switzerland) with CDCl₃ as the solvent. For the ¹H NMR spectrum, the relaxation delay is set to 3 s, with 128 scans performed. For the ²⁹Si NMR spectrum, the relaxation delay is set to 20 s with 256 scans completed.

The mechanical strength was determined using a universal tester (CMT 6000, SANS) with dumbbell-shaped specimens according to ASTM D638-2018. The size of the tensile sample is 50 mm × 4 mm × 2 mm, and the test was conducted at a cross-head speed of 50 mm/min.

The pencil hardness of the cured coatings was measured using a BY-500G pencil hardness tester (Shanghai, China), according to GB/T 6739-2006. The test was repeated five times at different locations on the sample surface to ensure the reliability and reproducibility of the results.

Contact angle measurements were performed with a JY-PHb apparatus (Chengde Jinhe Instrument Manufacturing Co., Ltd.). A microsyringe was employed to extrude 5 μL of deionized water or 2 μL of diiodomethane at the same rates

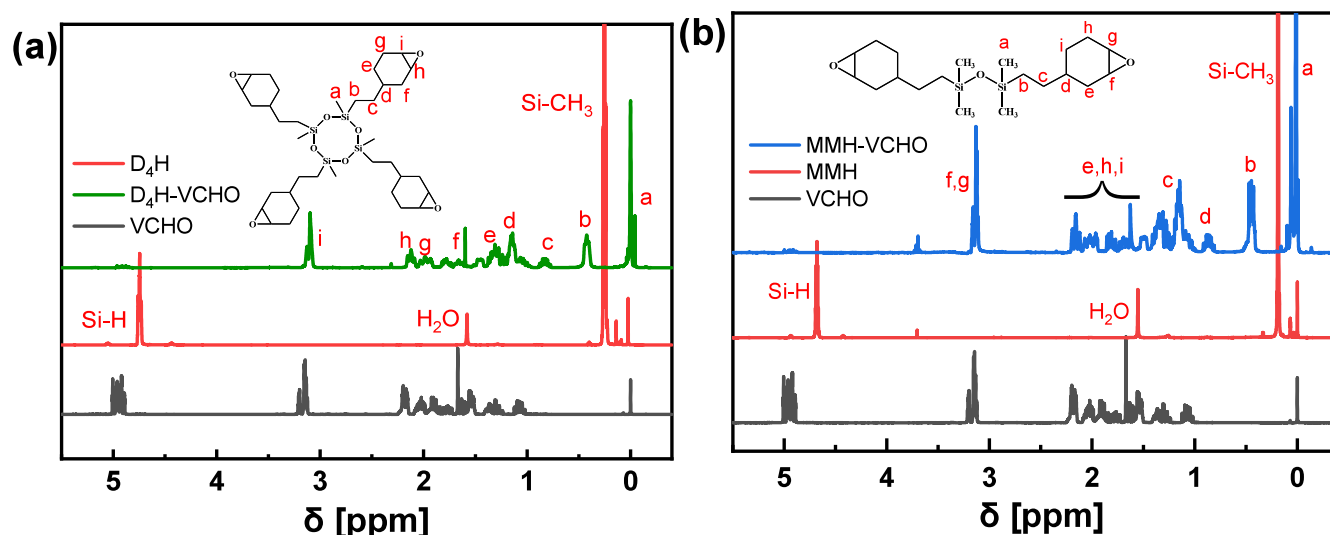


Figure 1. ^1H NMR spectra of D_4H , MMH, VCHO, and synthesized (a) $\text{D}_4\text{H-VCHO}$ and (b) MMH-VCHO in CDCl_3 .

onto the surface of the material to be examined. The contact angle was determined using the $\theta/2$ method of calibration, by documenting the data from five distinct areas of each material and averaging it as the contact angle of that substance.³² The surface energy of the material was calculated by the Owens–Wendt method based on the measured contact angles of deionized water and diiodomethane. The surface energy of the coating is calculated by eq 2,³³

$$\gamma_L \times (1 + \cos \theta) = 2(\gamma_S^d \times \gamma_L^d)^{1/2} + 2(\gamma_S^p \times \gamma_L^p)^{1/2} \quad (2)$$

Dynamic mechanical analysis (DMA) was carried with a DMA/DMA 50 (Metravib) instrument to measure the glass transition temperature, the storage module, and the loss angle of different samples. In this study, the samples were prepared in a film-like shape with dimensions of 10 mm $\times$ 5 mm $\times$ 0.1 mm, and a tensile mode was employed to evaluate their mechanical properties under tensile stress. The analysis was conducted with a displacement of 1×10^{-5} m and a frequency of 1 Hz. The samples were heated from 30 to 250 $^\circ\text{C}$ at a heating rate of 5 $^\circ\text{C}/\text{min}$.

The adhesion strength was determined by detaching the sample from the substrate, and the pull rate was at a speed of 0.2 MPa/s. Five different regions were tested for each sample to obtain an average value. In addition, the adhesion strength was measured again to characterize the underwater performance of the coating after soaking the coating in deionized water for 24 h.

The immersion test in a salt solution is conducted as follows: First, the tinplate sheet is evenly ground on both sides and cleaned with an acetone solution. After applying a protective coating to the tinplate, it is placed vertically in a 3.5 wt % NaCl solution. The setup is kept in a ventilated area, and the sodium chloride solution is replaced daily until the end of the test.

The corrosion behavior of the four coatings was assessed using EIS measurements performed on a CHI660E electrochemical workstation (Shanghai, China) at the open circuit potential (OCP). The coating thickness was 200 ± 5 μm , as determined by a CH-1-S thickness gauge (Shanghai Liuling Instrument Factory, China). Experiments were conducted in a 3.5 wt % NaCl solution with frequencies ranging from 10^{-2} to 10^5 Hz and a perturbation of ± 20 mV. Tests were carried out in a three-electrode cell where the coated sample served as the

working electrode with an exposed area of 3.14 cm^2 . All electrochemical measurements utilized a three-electrode system, comprising a saturated Ag/AgCl reference electrode, a graphite counter electrode, and Q235 steel (3.14 cm^2) as the working electrode. The EIS data were fitted by using the $Q(R(QR))$ model with ZSimpWin software.

Diatom counting was conducted by placing a drop of the diatom culture, after a period of incubation, onto a hemocytometer plate with an area of 1×1 mm^2 . The diatoms were then counted using an optical microscope equipped with a camera. Five central squares (upper left, lower left, upper right, lower right, and center) in the counting chamber of the hemocytometer were selected, and the total number of diatoms counted in these five squares was recorded as N_d . The diatom density (D_{cell}) was calculated using eq 3.¹¹

$$D_{\text{cell}} = N_d \div 80 \times 400 \quad (3)$$

where 5 large squares contain a total of 80 small squares, dividing by 80 gives the number of diatoms per small square. Multiplying this by 400 calculates the total number of diatoms in the counting chamber, and dividing by 1 square millimeter gives the diatom concentration in the solution.

The micromorphology of diatom cells remaining on the surface of the coating was observed by using a scanning electron microscope (SEM, Hitachi S-4800). Before observation, the samples were sputter-coated with a thin layer of gold to enhance the conductivity.

Bovine serum albumin (BSA) was chosen as the protein source for anti-protein adsorption test. 1 mL of BSA solution with a concentration of 5 mg/mL was applied to the coated surface in uniform drops and placed in a holding chamber at a constant temperature of 25 $^\circ\text{C}$ for 2 h of protein adsorption treatment. When the adsorption process was completed, the coating surface was again gently washed three times using a NaCl solution to remove the insufficiently adsorbed BSA, followed by blow-drying the coating surface using a stream of nitrogen gas. Subsequently, we added 1 mL of fluorescein isothiocyanate (FITC) (concentration of 0.143 mg/mL) and stained the BSA for 12 h at a low temperature of 0–4 $^\circ\text{C}$. At the end of staining, the sample surface was washed with methanol to remove residual fluorescein. Finally, the samples were placed under a fluorescence microscope for observation

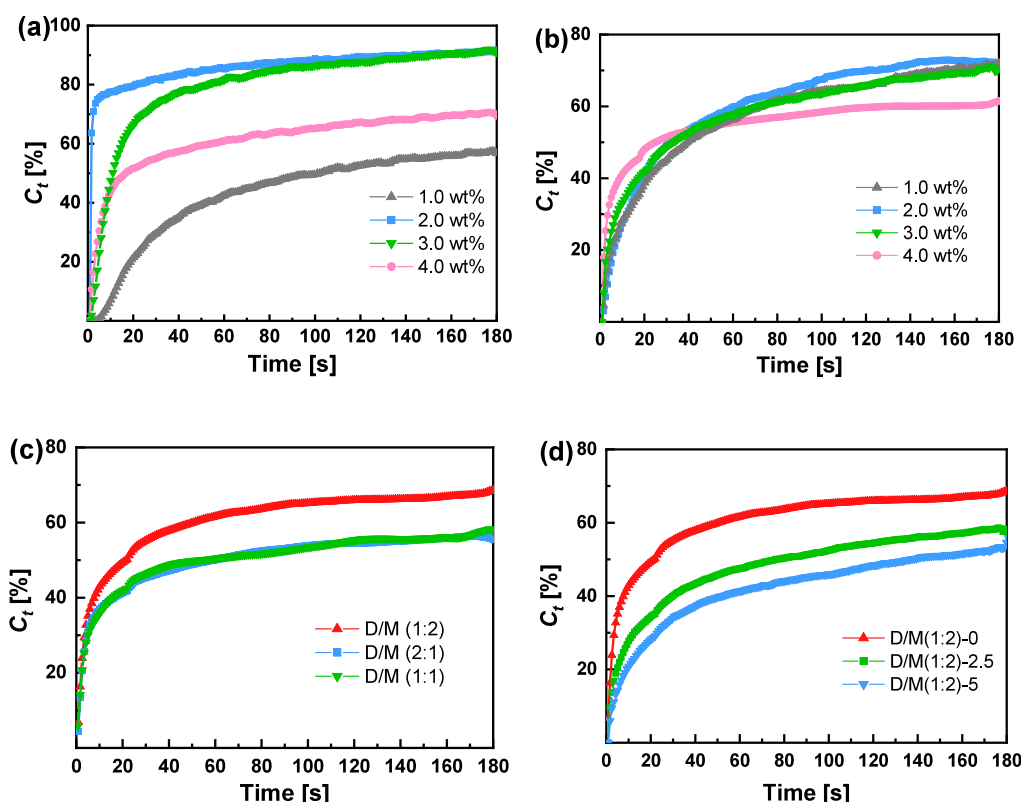


Figure 2. Relationship between the epoxy conversion rate and UV irradiation time is shown for (a) D₄H-VCHO and (b) MMH-VCHO with varying photoinitiator contents, (c) a mixture of D₄H-VCHO and MMH-VCHO at different mass ratios, and (d) a mixture of D₄H-VCHO and MMH-VCHO at a 1:2 ratio with varying Si-69 contents.

and photographed for recording.³⁴ The number of BSA proteins was counted under a fluorescence microscope and recorded as N_b . The number of BSA proteins in the blank sample was recorded as N_0 . The normalized protein adhesion rate, denoted as the RRR, was calculated using eq 4,

$$R = \frac{N_b}{N_0} \times 100\% \quad (4)$$

3. RESULTS AND DISCUSSION

3.1. Structural Characterization of D₄H-VCHO and MMH-VCHO. The structure of D₄H-VCHO and MMH-VCHO is confirmed through ¹H NMR, ²⁹Si NMR and FT-IR spectroscopy, as shown in Figure 1 and Figure S2. The ¹H NMR spectra of D₄H, MMH, VCHO, D₄H-VCHO, and MMH-VCHO are presented in Figure 1. The chemical shift between 0.40 and 2.15 ppm corresponds to the proton of 4-ethylcyclohexyl groups in VCHO, while the hydrogens on Si-CH₃ appear between 0 and 0.15 ppm in D₄H and MMH. On the curve of D₄H-VCHO and MMH-VCHO, the characteristic peak of Si-H at 4.7 ppm disappears, indicating that the hydrosilylation reaction proceeds completely. Additionally, the appearance of characteristic signal for the epoxide group and the 4-ethylcyclohexyl at 3.09 ppm and 0.4–2.15 ppm, respectively, suggests that cycloaliphatic epoxy groups have been successfully grafted onto the silicone molecules of D₄H and MMH.^{22,35} To investigate further, FT-IR spectrum is used to confirm the structure of D₄H-VCHO and MMH-VCHO. As shown in Figure S2a,b, the infrared peak located at 1640 cm⁻¹ represents the C=C bond of VCHO, and the infrared characteristic peaks for the Si-H bond in D₄H and MMH

appear at 2170 and 2134 cm⁻¹, respectively. On the curve of D₄H-VCHO and MMH-VCHO, however, the characteristic peaks of C=C bond and Si-H bond disappear, while the characteristic signals for Si-CH₃ and the epoxy group are observed at 1260 and 885 cm⁻¹, respectively.³⁶ Additionally, Figure S2c,d displays the ²⁹Si NMR spectra of D₄H-VCHO and MMH-VCHO, both of which exhibit unimodal peaks. The chemical shift at -19.93 ppm corresponds to (CH₃)₃Si-O in D₄H-VCHO, while the chemical shift at 7.56 ppm corresponds to (CH₃)₂Si-O in MMH-VCHO.²⁷ These results collectively confirm the successful synthesis of D₄H-VCHO and MMH-VCHO.

3.2. Basics of the Silicone-Epoxy Resin for UV-Curing.

To characterize the influence of varying amounts of cationic photoinitiator, the mass ratio between D₄H-VCHO and MMH-VCHO, and the content of Si-69 on the cationic photocuring rate of silicon-epoxy resins, the real-time infrared spectra (RT-IR) are used to measure the changes of epoxy groups over time. The mixture is dropped on KBr pellets and irradiated under UV light with a wavelength of 365 nm and an intensity of 40 mW/cm² for different time. Figure S3 shows the changes in the FT-IR spectra of the resin mixture of D₄H-VCHO and MMH-VCHO with a 1:2 mass ratio and 2 wt % cationic photoinitiator content under UV irradiation over time. During the cross-linking process, the epoxy groups react with each other, resulting in a gradual decrease in the intensity of the characteristic absorption peak at 883 cm⁻¹ as the UV irradiation time increases.³⁷ Accordingly, this study adopts the change in the integrated area of the 883 cm⁻¹ peak as a standard for evaluating the extent of the resin's photocuring reaction. Significantly, after UV irradiation, the peak of epoxy

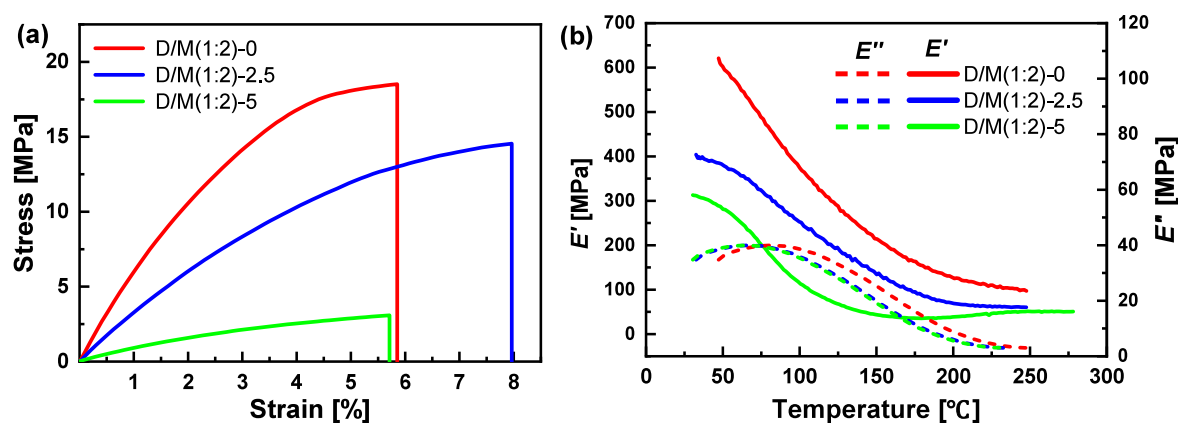


Figure 3. (a) Typical stress–strain curves and (b) the storage modulus and loss modulus curves of the UV-cured D/M(1:2)-0, D/M(1:2)-2.5 and D/M(1:2)-5 coatings.

groups does not disappear completely, even after prolonged irradiation, indicating incomplete conversion of the epoxy groups in the resin mixture.³⁸ This phenomenon is consistent with other reports, as achieving 100% conversion in photopolymers is inherently challenging.²²

In this study, the conversion rate of epoxy in the resins under UV irradiation is calculated using eq 1, where the value of C_t represents the degree of cross-linking, as shown in Figure 2. Specifically, Figure 2a,b shows the conversion rate curves of individual D₄H-VCHO and MMH-VCHO with varying amounts of cationic photoinitiator. It is evident that the conversion rate increases with UV irradiation time for both D₄H-VCHO and MMH-VCHO. Furthermore, the content of cationic photoinitiator has a more significant impact on the conversion rate of D₄H-VCHO compared to MMH-VCHO. Nonetheless, both compounds follow the same curing trend, where the conversion rate initially increases with the amount of photoinitiator but begins to decrease once the photoinitiator exceeds a certain level. The maximum conversion rate is observed at a photoinitiator content of 2%. Therefore, in subsequent studies, the photoinitiator is set at 2%.

Figure 2c displays the conversion rate of the resin mixture (D/M($n:m$)) composed of D₄H-VCHO and MMH-VCHO at three mass ratios: 1:2, 1:1, and 2:1. As illustrated in the figure, the D/M (1:2) mixture achieves the highest conversion rate, which can be attributed to its higher proportion of MMH-VCHO compared to the other mixtures. This may be because D₄H-VCHO serves as the cross-linker, and a higher content of D₄H-VCHO increases the difficulty of the curing process. Consequently, the D/M (1:2) mixture is selected to research the influence of Si-69 on the conversion rate. D/M (1:2)-0, D/M (1:2)-2.5, and D/M (1:2)-5 represent mixtures where the amount of Si-69 is 0%, 2.5%, and 5% of the total weight of D₄H-VCHO and MMH-VCHO, respectively. As shown in Figure 2d, the conversion rates in both D/M (1:2)-0 and D/M (1:2)-2.5 exceed 50% when the reaction time surpasses 80 s. Obviously, the addition of Si-69 significantly reduces both the conversion rate and the overall conversion degree of epoxy groups. This effect occurs because Si-69 does not contain photocuring reactive groups and remains as free small molecules in the system, which interferes with the effectiveness of the cationic photoinitiator.³⁹

3.3. Mechanical Properties of the UV-Cured Silicone-Epoxy. The mechanical properties of a coating are critical parameters that determine its strength, durability, and

flexibility. Figure S4a presents the stress–strain curves of UV-cured coatings for individual D₄H-VCHO and MMH-VCHO. The results reveal that the tensile strength and elongation at break of D₄H-VCHO-cured are lower than those of MMH-VCHO-cured, which is attributed to the excessively fast UV-curing rate of D₄H-VCHO. Before the UV-curing process, the resin mixture is sprayed onto a metal substrate and irradiated with UV light. During the UV-curing process, the solvent rapidly evaporates, and the excessively fast curing rate prevents the timely relief of shrinkage stress and leads to the formation of minute flaws or fissures in the coating, making it highly susceptible to cracking and resulting in reduced mechanical properties. However, the individual MMH-VCHO resin undergoes significant volume shrinkage during the curing process, which results in the formation of numerous wrinkles on the surface, as shown in Figure S5. To enhance the performance of the coating, MMH-VCHO is added to D₄H-VCHO as flexible bridges owing to its two epoxy groups and flexible silicone chain. The typical stress–strain curves of the UV-cured D/M(1:2), D/M(1:1), and D/M(2:1) resins are shown in Figure S4a, with detailed mechanical properties listed in Table S1. As described, the tensile strength of D/M(1:2), D/M(1:1), and D/M(2:1) is 18.5 MPa, 19.0 MPa, and 15.5 MPa, respectively, maintaining a relative high level compared with conventional silicone coating.^{40–42} Notably, the elongation at break for D/M(1:2) is 5.8%, which is higher than those of D/M(1:1) and D/M(2:1). This can be attributed to the higher proportion of MMH-VCHO in the coating, resulting in lower internal stress and improved toughness. Therefore, a 1:2 mass ratio of D₄H-VCHO to MMH-VCHO is identified as the optimal proportion for the coating.

Figure 3 presents the stress–strain curves of UV-cured D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings, illustrating the effect of Si-69 content on the mechanical properties. The results reveal that the tensile strength of the coatings decreases with increasing Si-69 concentration, with values of 18.5, 14.5, and 3.1 MPa, respectively. Furthermore, the elongation at break for D/M(1:2)-2.5 is higher than that for D/M(1:2)-0 and D/M(1:2)-5. It is worth noting that the storage modulus (E'), loss modulus (E'') and glass transition temperatures (T_g) of the three coatings, as measured by DMA, are also critical mechanical performance parameters of the coatings. Figures 3b and S4b show the relationship between E' , E'' , and T_g with temperature for the UV-cured D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings. The results reveal

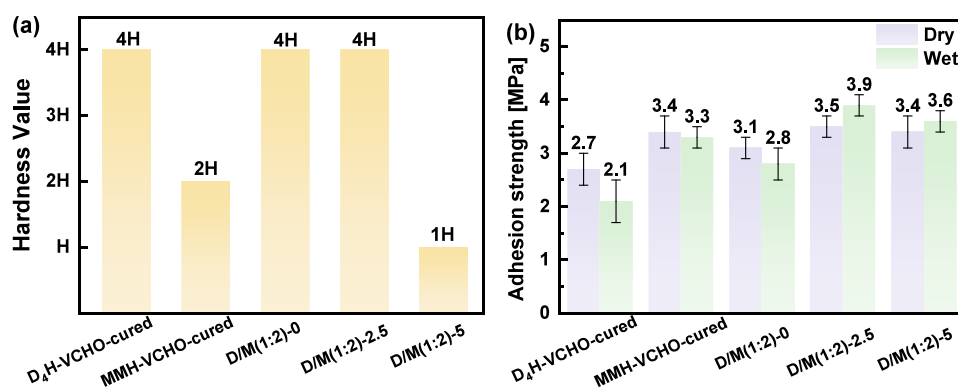


Figure 4. (a) The hardness of the coatings and (b) the adhesion strengths of the UV-cured samples coated between tinplate and glass, both dry and after immersion in water for 24 h.

that the value of E' decreases with the addition of Si-69, while the T_g for D/M(1:2)-2.5 is the highest, reaching 155 °C. This trend is consistent with the observations from the stress–strain curves. We speculate that the decrease in tensile strength and E' is caused by the introduction of Si-69, which reduces the reaction between MMH-VCHO and D₄H-VCHO, leading to a decrease in cross-linking density. To quantitatively characterize the effect of Si-69 content on the reaction between MMH-VCHO and D₄H-VCHO, the cross-linking density of the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings is calculated by eq 5:⁴³

$$\nu_e = \frac{E'}{6RT} \quad (5)$$

where ν_e is the cross-linked density, E' is the storage modulus of rubber platform which can be measured at $T_g + 40$ °C, R is the universal gas constant, T is the temperature corresponding to E' . The values of ν_e for the UV-cured D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings are summarized in Table S2 as 6872.1, 3155.9, and 1777.7 mol/m³, respectively. These results confirm that a higher Si-69 content in the resin leads to a lower cross-linking density in the UV-cured coatings. This is because Si-69 reduces the activity of the photoinitiator. Furthermore, when the addition of Si-69 is 2.5%, both the elongation at break and the T_g reach their highest values. This is because a small amount of Si-69 increases the free volume and the polarity of the cross-linking network, thereby improving the toughness and the intermolecular force within the network. However, when the Si-69 concentration exceeds 2.5%, the mechanical properties of the coating deteriorate because Si-69 acts as plasticizer in the UV-cured coatings, loosening the cross-linked network. Overall, 2.5% is identified as the optimal Si-69 addition level at which the coating achieves the best balance between the photocuring conversion rate and mechanical properties.

3.4. Physicochemical Characteristics of the UV-Cured Coating. The hardness of a coating is a crucial property that determines its resistance to scratches and mechanical wear, significantly impacting its durability and overall performance. The scratch resistance of the UV-cured coating is measured using a BY-500G pencil hardness tester. Figure 4a shows the hardness values of D₄H-VCHO-cured, MMH-VCHO-cured, D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings. It is evident that the hardness of the UV-cured coating prepared in this study is directly related to the content of D₄H-VCHO and Si-69. The D₄H-VCHO-cured coating exhibits twice the

hardness of the MMH-VCHO-cured coating, which can be attributed to the rigid Si–O four-membered ring structure of D₄H-VCHO and the highly cross-linked network formed by its tetrafunctional groups. In the mixed coatings D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5, the scratch resistance primarily originates from the D₄H-VCHO component. As mentioned in the previous section, an excessive amount of Si-69 significantly reduces the cross-linking density of the coating. As a result, when the cross-linking density drops below a certain threshold, the scratch resistance of the coating is also greatly reduced. Obviously, the optimal Si-69 content is 2.5%, at which the hardness value of D/M(1:2)-2.5 reaches 4 H, a level comparable to that of ceramic.^{44–46}

The adhesion strength between the coating and substrate plays a pivotal role in determining the effectiveness, durability, and long-term performance of the resins. The adhesive strengths of the UV-cured coatings, including D₄H-VCHO-cured, MMH-VCHO-cured, D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5, are tested on tinplate under shear conditions at five different positions for each coating, and the average value is calculated to evaluate their performance in both dry and wet environments. As described in Figure 4b, the adhesive strength of MMH-VCHO-cured is higher than that of D₄H-VCHO-cured under both dry and wet conditions. This superiority is attributed to MMH-VCHO-cured coating having fewer flaws compared to D₄H-VCHO-cured, as the excessively fast curing rate of D₄H-VCHO leads to the formation of more defects in the coating. Moreover, the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings consistently demonstrated high adhesive strengths under dry conditions, measuring 3.1, 3.5, and 3.4 MPa, respectively. The addition of Si-69 did not decrease the adhesive strengths between the coatings and the substrate; instead, it slightly increased them. This enhancement is due to the formation of silicon–oxygen–metal (Si–O–M) bonds between Si-69 and the metal surface, enhancing the interaction between the silicon-epoxy resin and the substrate.^{47–49} Remarkably, after being soaked in water for 24 h, the adhesion strength of D/M(1:2)-0 decreased, while the adhesion strengths of D/M(1:2)-2.5 and D/M(1:2)-5 actually increased, from 3.5 MPa and 3.4 MPa to 3.9 MPa and 3.6 MPa, respectively. This increase is attributed to the hydrolysis of Si-69 during the immersion process, which forms a siloxane network (Si–O–Si) and facilitates interface cross-linking with the silicon-epoxy resin.

For the UV-curable fouling release coatings (FRCs) developed in this study, the contact angle and surface free

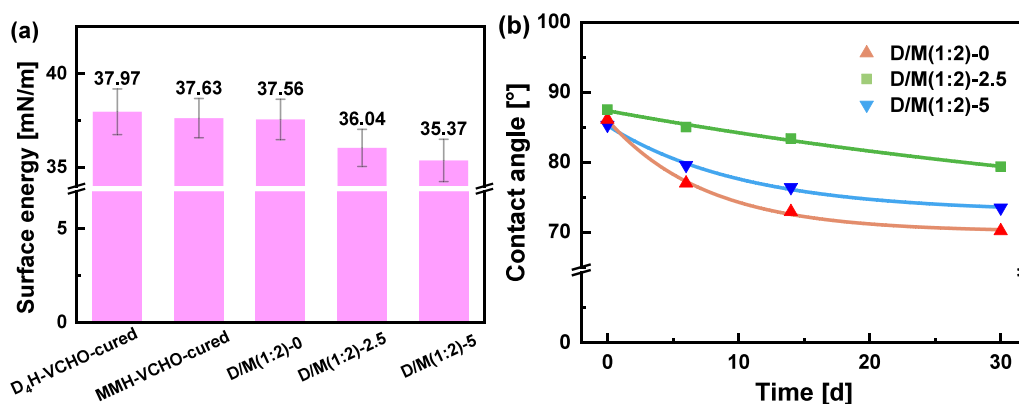


Figure 5. (a) Surface energy of the UV-cured coating. (b) Contact angle of the UV-cured coating as a function of immersion time in deionized water.

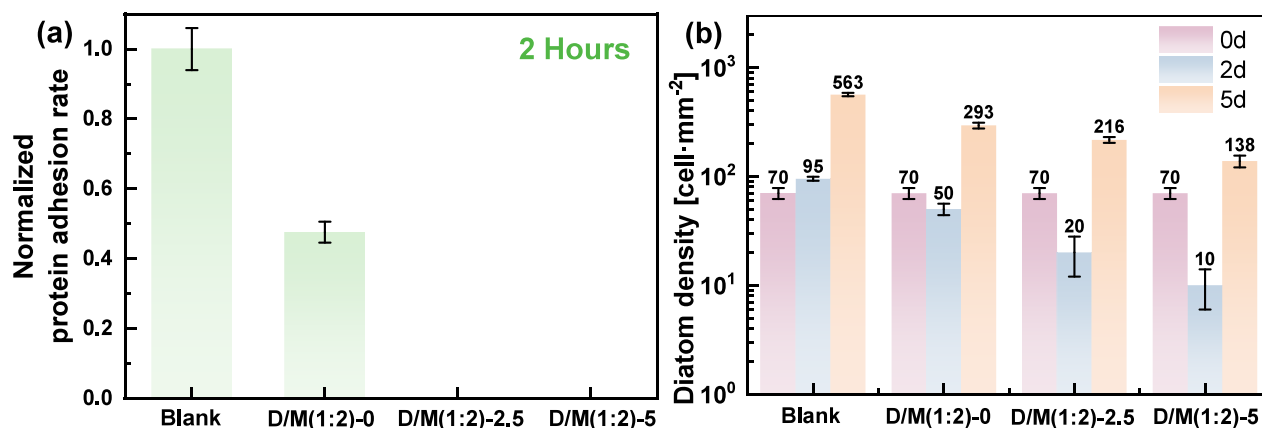


Figure 6. Antifouling abilities of different coatings: the uncoated blank sample, D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5. (a) The protein adhesion rate after BSA was applied on the surface of the samples for 2 h. (b) The adhesion density of diatom after soaking in a Chlorella culture solution for 0, 2, and 5 days.

energy (SFE) are critical factors that influence their antifouling and detachment properties. Typically, the lower the SFE of a coating, the stronger is its antibiofouling ability; the smaller the variation in contact angle over immersion time in deionized water, the better is the coating's stability. The contact angles of D₄H-VCHO-cured, MMH-VCHO-cured, D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings with water and diiodomethane are measured. For each coating, the contact angles are tested at five different positions and the average values are calculated. The contact angle results are shown in Figure S6. Subsequently, the SFE of the coatings is calculated by using eq 2, with the SFE value presented in Figure 5a. Studies indicate that the SFE of traditional silicone coatings ranges from 20 to 30 mN/m, while that of epoxy coatings is between 44.4 and 59.5 mN/m.⁵⁰ The UV-cured coatings developed in this study, which incorporate both polar epoxy groups and low surface energy Si–O bonds, exhibit an SFE ranging from 35 to 40 mN/m, positioning them between traditional silicone and epoxy coatings in terms of surface free energy. Moreover, Figure 5a demonstrates that the SFE values of D/M(1:2)-x coatings decrease with increasing Si-69 content. This reduction in SFE is attributed to the synergistic effect of two factors caused by the increase in Si-69 content: the higher concentration of Si–O bonds in the network and the reduced surface irregularities resulting from a decrease in cross-linking density, both contributing to the lower SFE values.⁵¹

The contact angle stability of a coating in water is critical for achieving long-term antifouling and anticorrosion performance. To evaluate the effect of Si-69 on the stability of the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings, the contact angles of these coatings are further examined after a period of water immersion. As shown in Figure 5b, the contact angles of the three coatings are similar before immersion in deionized water, ranging from 85° to 88°. During the initial stages of the soaking, the contact angles slightly decrease due to the rearrangement of the polymer molecular chain on the surface of the coating when in contact with water. Over time, the contact angles tended to stabilize. It is evident that the decrease in contact angle is smallest for the D/M(1:2)-2.5 coating as soaking time increases. This stability can be attributed to the formation of a siloxane monolayer by the hydrolysis of Si-69 on the surface, which prevents water permeation. Furthermore, D/M(1:2)-2.5 coating exhibits a more uniform surface compared to D/M(1:2)-5, due to its higher cross-linking density. Together, these factors contribute to the prolonged structural stability of the D/M(1:2)-2.5 coating.

3.5. Antifouling and Anticorrosion Properties. In this study, the silane coupling agent Si-69, containing tetrasulfide (S–S–S–S) bonds, is introduced into UV-cured silicone-epoxy coatings to enhance their antifouling ability by leveraging the excellent antibacterial activity of the S–S–S–S bonds. To comprehensively evaluate the effect of Si-69 on

the coatings' resistance to biological fouling, bovine serum albumin (BSA) and *Chlorella* are used as fouling organism models for qualitative and quantitative characterization.

In the initial stages of the marine fouling process, proteins act as a primary nutrient source for the growth of bacteria and other microorganisms. Hence, a lower amount of BSA adsorbed on the coating surface indicates a stronger resistance to biofouling.⁵² To quantitatively evaluate the ability of S–S–S bonds to prevent protein adhesion, an uncoated tinplate is used as the blank control sample. The protein adhesion rates of the D/M (1:2)-0, D/M (1:2)-2.5, and D/M (1:2)-5 coatings are calculated using eq 4. Figure S7 shows microscopic images of the sample surfaces after 2 h of BSA adsorption, while the Figure 6a provides a comparative analysis of protein adhesion rates, calculated based on the number of proteins observed in Figure S7. The results reveal that, compared to the uncoated blank control, the protein adhesion rate on the surface of the D/M(1:2)-0 coating is reduced by half, reaching 0.46, while the surfaces of D/M(1:2)-2.5 and D/M(1:2)-5 coatings exhibit almost no BSA adsorption. This reduction in the D/M (1:2)-0 coating is attributed to the presence of Si–O bonds, which lower the surface free energy and make BSA adsorption more difficult, thereby decreasing the protein adhesion rate. Furthermore, the D/M (1:2)-2.5 and D/M (1:2)-5 coatings benefit from the inclusion of Si-69, which interacts with BSA through exchange reactions between the S–S–S–S bonds and the thiol (–SH) groups in the protein. This interaction disrupts the conformational integrity of BSA, significantly reducing its ability to adsorb onto the coating surface.

In the second stage of marine fouling process, diatoms begin to form a biofilm on the subsea surface of equipment, creating favorable conditions for the attachment of other organisms.⁵³ To evaluate the antidiatom adhesion ability of the UV-cured silicon-epoxy coatings, *Chlorella* cells are selected as the test model. Figures 6b and 7 show the qualitative and quantitative

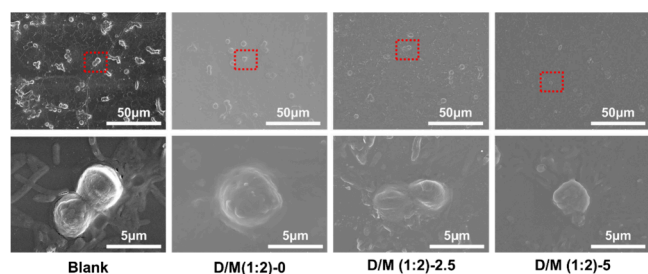


Figure 7. Diatom attachment on the surface of tested tinplate samples for 5 days.

results of the coatings' antidiatom performance. Figure 6b illustrates the relationship between the density of diatoms on the coating surfaces and the soaking time in a *Chlorella* solution. The adhesion density is calculated using eq 3. Figure 7 displays an SEM image of the surfaces of the uncoated blank sample and D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings after soaking in *Chlorella* solution for 5 days. As described in Figure 6b, after soaking in *Chlorella* solution for 2 days, the diatom density on the surface of the blank control sample increases, while the diatom density on the surfaces of the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings decreases, with greater reductions observed for coatings with higher Si-69 content. After 5 days of soaking, the diatom density on all sample surfaces increases significantly; however,

coatings with a higher Si-69 content exhibit lower diatom densities. This trend aligns with the observations in Figure 7, where the amount of diatom adhesion on the coating surfaces decreases as the Si-69 content increases. Notably, the diatom cells on the surfaces of the D/M(1:2)-2.5 and D/M(1:2)-5 coatings exhibit significant rupture compared to those on the D/M(1:2)-0 coating. This effect can be attributed to the S–S–S–S bonds in Si-69, which disrupt the biological activity of diatom cells by damaging their cell membrane metabolism.

To provide a clearer demonstration of the corrosion resistance of the UV-cured silicon-epoxy coatings, the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings are continuously immersed in 3.5 wt % NaCl solution for 10 and 21 days. The corrosion state of the tinplate substrates is shown in Figure 8. After 21 days of continuous immersion, the

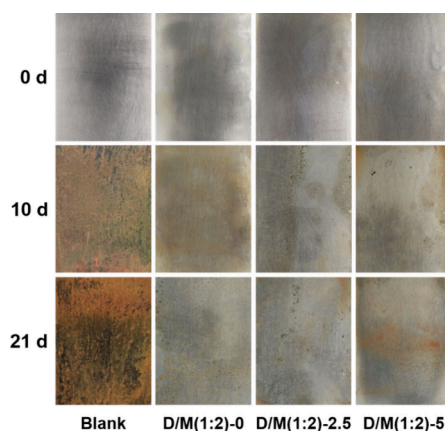


Figure 8. Corrosion status of tested tinplate samples (uncoated and coated with D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5) after immersion in a 3.5 wt % NaCl solution for 0, 10, and 21 days.

surface of the tinplate in the uncoated blank control sample exhibits severe corrosion, while the tinplates protected by the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings exhibited only mild corrosion. Additionally, the corrosion resistance of D/M(1:2)-0 and D/M(1:2)-2.5 is comparable, and both outperform D/M(1:2)-5. This difference can be attributed to the lower mechanical properties and hardness of the D/M (1:2)-5 coating, which allow the NaCl solution to penetrate more easily, leading to substrate corrosion.

Electrochemical impedance spectroscopy (EIS) is an effective method for assessing the barrier performance and corrosion resistance of coatings.^{54,55} Figure 9 displays the Nyquist and Bode plots of tinplate plates coated with UV-cured protective layers, specifically D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5, after immersion in a 3.5 wt % NaCl solution for 3 days and 21 days. In general, the radius of the capacitive loop in Nyquist plots is an important indicator for evaluating the protective performance of coatings, with a larger radius under identical conditions indicating stronger corrosion resistance.⁵⁶ Similarly, in Bode plots, the impedance modulus at the lowest frequency ($|Z|_{0.01 \text{ Hz}}$) serves as a semiquantitative parameter to assess the barrier properties of the coating against corrosive media.⁵⁷ Due to the very high impedance values of the coatings in this study, only a half semicircle was observed in the Nyquist plots for all samples.

As shown in Figure 9a,c, the capacitive loops in the Nyquist plots for the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings shrink significantly as the immersion time in 3.5 wt %

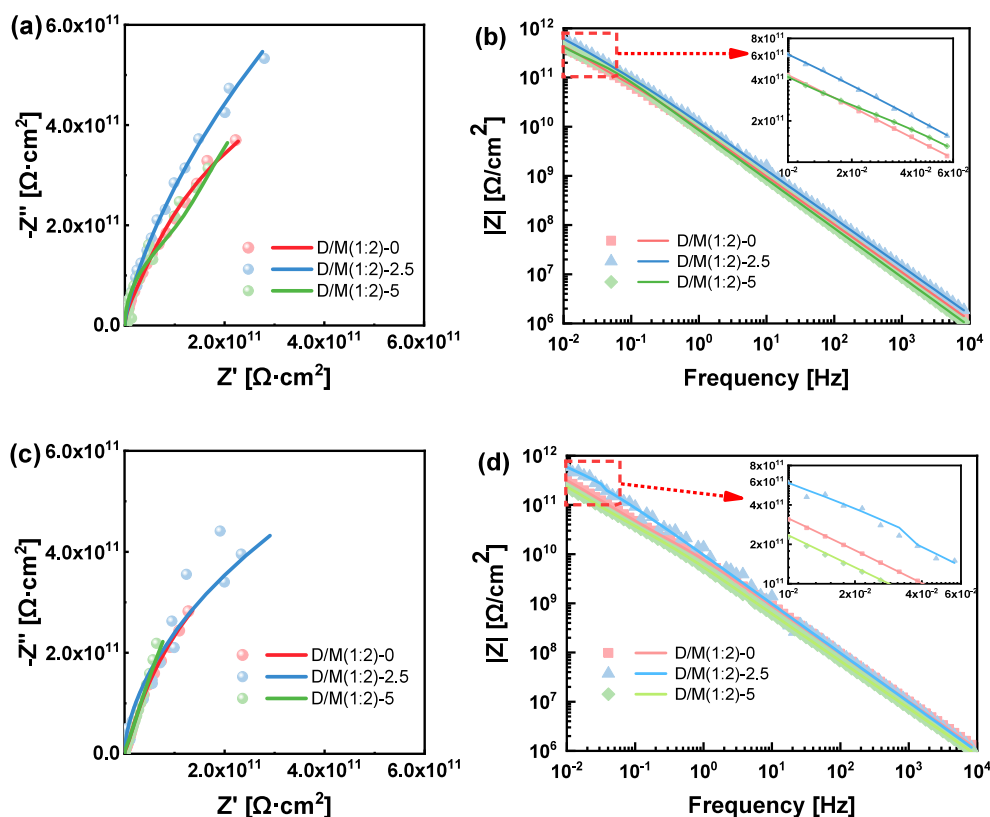


Figure 9. Nyquist (a, c) and Bode (b, d) plots of three coatings with different Si-69 content immersed in a 3.5 wt % NaCl solution for 3 days (a, b) and 21 days (c, d).

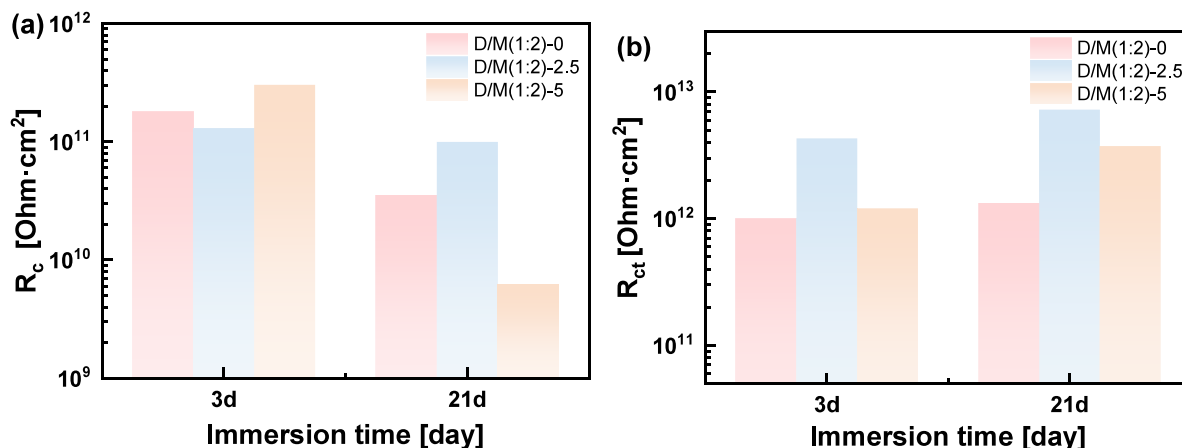


Figure 10. (a) R_c and (b) R_{ct} of D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings at various immersion times.

NaCl solution increases from 3 to 21 days, indicating a noticeable reduction in the radius of the impedance arcs. This suggests a decline in corrosion resistance, implying that the anticorrosion performance of all three coatings decreases over time. However, at any given immersion time, the Nyquist plots show that the capacitive loop radius of the D/M(1:2)-2.5 coating is consistently larger than those of the other two coatings, indicating that the D/M(1:2)-2.5 coating exhibits superior corrosion resistance compared with the D/M(1:2)-0 and D/M(1:2)-5 coatings. Additionally, Figure 9b,d reveals that the Bode impedance curves of the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings remain nearly linear, with $|Z|_{0.01 \text{ Hz}}$ values exceeding $10^{11} \Omega \cdot \text{cm}^2$ after both 3 and 21 days of immersion. This demonstrates that the silicon-epoxy

coatings developed in this study effectively inhibit the penetration of corrosive media, maintaining their structural integrity even during prolonged immersion and providing durable corrosion protection.^{58,59} It is worth noting that the impedance modulus exhibits a markedly different trend compared to that of the capacitive loop. While the capacitive loop radius decreases with increasing immersion time (indicating a reduction in the activation energy of the corrosion reaction), the $|Z|_{0.01 \text{ Hz}}$ values remain stable. This phenomenon suggests that although the interaction between the corrosive medium and the coating surface intensifies over time, the corrosive species have not fully penetrated the coatings. The stable impedance modulus at low frequencies indicates that the inner structure of the coatings remains intact,

ensuring the continued protection of the underlying substrate. The shrinkage of the capacitive loop radius over time can be attributed to processes such as water uptake, chloride ion penetration, or microstructural changes at the coating surface caused by extended exposure to the aggressive chloride-containing environment, which reduce the coating's resistance to corrosion while leaving its inner barrier function unaffected.

The equivalent circuit serves as an effective tool for analyzing EIS data, enabling a quantitative evaluation of the corrosion resistance of the coatings. The corrosion resistance of the D/M(1:2)-0, D/M(1:2)-2.5, and D/M(1:2)-5 coatings is evaluated by fitting their EIS data to the equivalent circuit model shown in Figure S8. The fitting results are displayed in Figure 10 and Table S3. R_c represents the pore resistance of the coating, where a higher value indicates stronger barrier properties against corrosive media. R_{ct} represents the charge transfer resistance at the interface, with a higher value corresponding to a slower corrosion rate.⁶⁰ As shown in Figure 10a, at the beginning of the immersion in a 3.5 wt % NaCl solution, the R_c values of the three coatings are similar, with the R_c value of the D/M(1:2)-2.5 coating being the lowest. However, after 21 days of immersion, the R_c value of the D/M(1:2)-2.5 coating becomes significantly higher than those of the D/M(1:2)-0 and D/M(1:2)-5 coatings. Moreover, throughout the immersion period, the R_{ct} values of the D/M(1:2)-2.5 coating remain consistently higher than those of the D/M(1:2)-0 and D/M(1:2)-5 coatings, as shown in Figure 10b. These findings demonstrate that the D/M(1:2)-2.5 coating exhibits the most effective anticorrosion performance. This superior performance can be attributed to its excellent mechanical properties, coupled with the formation of a siloxane monolayer from Si-69, which synergistically prevents water intrusion and enhances the coating's anticorrosion capabilities.

4. CONCLUSIONS

Inspired by the antimicrobial mechanism of polysulfide bonds in GO, this study introduces Si-69, featuring S–S–S–S bonds, into a cycloaliphatic silicone-epoxy resin mixture of D₄H-VCHO and MMH-VCHO. The result is a UV-curable organosilicon coating with excellent antifouling and anticorrosive capabilities. The cross-linking of D₄H-VCHO and MMH-VCHO enables precise tuning of the coating's mechanical properties without compromising its surface free energy. Additionally, the S–S–S–S bonds enhance the antifouling performance by disrupting the biological activity of cell membranes. This approach successfully achieves a single organosilicon coating that integrates two opposing properties: strong adhesion to the substrate and low adhesion to marine organisms. Comprehensive analysis shows that the curing rate, mechanical properties, hardness, antifouling, and anticorrosive effects of the coating can be effectively tuned by adjusting the ratios of D₄H-VCHO, MMH-VCHO, and Si-69. Optimal results are obtained when the mass ratio of D₄H-VCHO to MMH-VCHO is 1:2 and the content of Si-69 is 2.5%. The coating with the specific formulation exhibits a tensile strength of 14.5 MPa, a hardness of 4 H, and dry and wet adhesion strengths to a tinplate substrate of 3.5 MPa and 3.9 MPa, respectively. It also demonstrates excellent inhibition of protein and diatom adhesion as well as anticorrosive properties. This mechanically robust, multifunctional organosilicon coating offers great potential for marine antifouling

applications, addressing the critical need for environmentally friendly and effective solutions.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.4c04473>.

Additional figures showing structures of D₄H-VCHO, MMH-VCHO, and silicone-epoxy coatings; FT-IR spectra; ²⁹Si NMR spectra; real-time FT-IR spectra; stress–strain curves; loss factor curves; contact angles; coating surface; protein adsorption surface; and electrical equivalent circuit models; additional tables listing mechanical properties, cross-linked density, and EIS fitting results for different samples (PDF)

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Author Contributions

Junjie Huang: writing—original draft, methodology, investigation, formal analysis. Yuan Cao: writing—review and editing, validation, supervision, methodology, conceptualization. Hongping Xiang: conceptualization, resources, funding acquisition.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grant 52273104), Guangdong Basic and Applied Basic Research Foundation (Grant 2022A15-15011972), and the National Postdoctoral Researcher Funding Program (Grant GZC20230562). The authors thank Chen Yu and Yu Ling from the Guangdong University of Technology analysis and test center for their assistance with NMR spectroscopy.

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